

3.1 *Theorem:*

$$\langle p | \hat{T} = \exp\left(\frac{-ibp}{\hbar}\right) \langle p |.$$

Proof. Given

$$\langle x | \hat{T} = \langle x - b |, \quad (1)$$

we can expand with

$$\begin{aligned} \langle p | \hat{T} &= \sum_x \langle p | x \rangle \langle x | \hat{T} \\ &= \int_{-\infty}^{\infty} \exp\left(\frac{-ipx}{\hbar}\right) \langle x - b | dx \end{aligned}$$

and if we let $\xi = x - b$ and subsequently $x = \xi + b$ and $dx = d\xi$,

$$\begin{aligned} \langle p | \hat{T} &= \int_{-\infty}^{\infty} \exp\left(\frac{-ip(\xi + b)}{\hbar}\right) \langle \xi | d\xi \\ &= \int_{-\infty}^{\infty} \exp\left(\frac{-ip\xi}{\hbar}\right) \exp\left(\frac{-ipb}{\hbar}\right) \langle \xi | d\xi \\ &= \exp\left(\frac{-ipb}{\hbar}\right) \int_{-\infty}^{\infty} \exp\left(\frac{-ip\xi}{\hbar}\right) \langle \xi | d\xi \end{aligned}$$

and the second term is an equivalent definition of $\langle p |$ as ξ lies in position-space, simplifying to

$$\langle p | \hat{T} = \exp\left(\frac{-ipb}{\hbar}\right) \langle p |.$$

□

3.2 *Theorem:*

$$\langle p | \hat{D} = \frac{ip}{\hbar} \langle p |.$$

Proof. Given the definition

$$\langle x | \hat{D} = \frac{d}{dx} \langle x |, \quad (2)$$

we can once again expand

$$\begin{aligned} \langle p | \hat{D} &= \sum_x \langle p | x \rangle \langle x | \hat{D} \\ &= \int_{-\infty}^{\infty} \exp\left(\frac{-ipx}{\hbar}\right) \frac{d}{dx} \langle x | dx \end{aligned}$$

Using integration by parts, we can set

$$\begin{aligned} u &= \exp\left(-\frac{ipx}{\hbar}\right) & du &= -\frac{ip}{\hbar} \exp\left(-\frac{ipx}{\hbar}\right) dx \\ dv &= \frac{d}{dx} \langle x | dx & v &= \langle x | \end{aligned}$$

yielding the equation

$$\begin{aligned} \int_{-\infty}^{\infty} \exp\left(-\frac{ipx}{\hbar}\right) \frac{d}{dx} \langle x | dx &= \exp\left(-\frac{ipx}{\hbar}\right) \langle x | \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} -\frac{ip}{\hbar} \exp\left(-\frac{ipx}{\hbar}\right) \langle x | dx \\ &= \exp\left(-\frac{ipx}{\hbar}\right) \langle x | \Big|_{-\infty}^{\infty} + \frac{ip}{\hbar} \int_{-\infty}^{\infty} \exp\left(-\frac{ipx}{\hbar}\right) \langle x | dx \end{aligned}$$

As $\lim_{x \rightarrow \pm\infty} \exp(-ipx/\hbar) = 0$, both terms simplify to

$$\langle p | \hat{D} = \frac{ip}{\hbar} \langle p |.$$

□

3.3 The mirror reflection operator $\hat{R}$ produces the reflection about the origin of any function in x -space. It is called the parity operator in physics. Its x -space instruction is $\langle x | \hat{R} = \langle -x |$. What is its p -space instruction?

Suppose we have a state ψ ,

$$\begin{aligned} \langle p | \hat{R} | \psi \rangle &= \sum_x \langle p | x \rangle \langle x | \hat{R} | \psi \rangle \\ &= \int_{-\infty}^{\infty} \exp\left(-\frac{ipx}{\hbar}\right) \langle -x | \psi \rangle dx \end{aligned}$$

Now let $\xi = -x$, producing

$$\begin{aligned} \langle p | \hat{R} | \psi \rangle &= - \int_{-\infty}^{\infty} \exp\left(\frac{ip\xi}{\hbar}\right) \langle \xi | \psi \rangle d\xi \\ &= - \int_{\infty}^{-\infty} \exp\left(-\frac{i(-p)\xi}{\hbar}\right) \langle \xi | \psi \rangle d\xi \\ &= \langle -p | \psi \rangle \end{aligned}$$

3.4-1 Given $\hat{\tau} |p\rangle = |p + Q\rangle$ where $Q = \hbar/L$. *Theorem:*

$$\langle p | \hat{\tau} = \langle p - Q |.$$

Proof. $\langle p | \hat{\tau} = (\hat{\tau} |p\rangle)^* = \langle p | \hat{\tau}^\dagger = \langle p - Q |$ □

3.4-2 *Theorem:*

$$\langle x | \hat{\tau} = \exp\left(\frac{iQx}{\hbar}\right) \langle x |$$

Proof.

$$\begin{aligned}\langle x | \hat{\tau} &= \sum_p \langle x | p \rangle \langle p | \hat{\tau} = \sum_p (\langle p | x \rangle)^* \langle p | \hat{\tau} \\ &= \sum_p \exp\left(\frac{ipx}{\hbar}\right) \langle p - Q | \end{aligned}$$

Now let $p' = p - Q$,

$$\begin{aligned}\langle x | \hat{\tau} &= \sum_p \exp\left(\frac{i(p' + Q)x}{\hbar}\right) \langle p' | \\ &= \sum_p \exp\left(\frac{ip'x}{\hbar}\right) \exp\left(\frac{iQx}{\hbar}\right) \langle p' | \\ &= \exp\left(\frac{iQx}{\hbar}\right) \sum_p \exp\left(\frac{ip'x}{\hbar}\right) \langle p' | \\ &= \exp\left(\frac{iQx}{\hbar}\right) \langle x | \end{aligned}$$

□

3.5-1 *Theorem:*

$$\langle x | \hat{\tau} \hat{D} | \psi \rangle = \exp\left(\frac{iQx}{\hbar}\right) \frac{d\psi}{dx}$$

Proof. $\langle x | \hat{\tau} \hat{D} | \psi \rangle = \frac{d}{dx} \langle x | \hat{\tau} | \psi \rangle = \exp\left(\frac{iQx}{\hbar}\right) \frac{d}{dx} \langle x | \psi \rangle.$

□

7.1 *Theorem:* $\hat{T} = e^{-b\hat{D}}$

Proof.

$$\langle p | \exp(-b\hat{D}) = \langle p | \exp(-bD(p))$$

where $D(p) = ip/\hbar$ is the eigenvalue of $\hat{D}$ for $\langle p |$, and knowing that

$$\langle p | \hat{T} = \exp\left(\frac{-ibp}{\hbar}\right) \langle p | \quad (3)$$

from problem 3.1,

$$\langle p | \hat{T} = \exp(-bD(p)) \langle p | = \exp(-b\hat{D}) \langle p |,$$

thus

$$\hat{T} = e^{-b\hat{D}}.$$

□

7.2 *Theorem:* $\langle x|\hat{p}|\psi\rangle = \langle x|\hat{p}\hat{T}|\psi\rangle$

Proof. Given that $\hat{T} = e^{-b\hat{D}}$ per problem 7.1, $\langle x|\psi\rangle$ and $\langle x|\hat{T}|\psi\rangle$ differ by only the phase $e^{-b\hat{D}}$ which cannot have any effect on the probability as $\forall\theta, |e^{i\theta}|^2 = 1$. $\square$

7.3 *Theorem:* $\forall\psi, \forall t, |\langle E|\psi\rangle|^2 = |\langle E|\hat{U}_t|\psi\rangle|^2$.

Proof. $\hat{U}_t = \exp\left(\frac{-it\hat{H}}{\hbar}\right)$ is a unit vector on the complex plane, so it cannot have any effect on the probability. $\square$

7.4 What is the characteristic frequency $\Omega/2\pi$ for a neutron in a magnetic field of 1 kG? In what region of the electromagnetic spectrum is this: radio, microwave, infrared, visible, or x-ray?

The characteristic frequency is

$$\omega = \frac{B}{2\pi}|\Omega| = \frac{B}{2\pi}\left|\frac{g_n e}{2m_n}\right| = \frac{1 \times 10^3 \text{ kG}}{2\pi} \left|\frac{(-3.83)(1.60 \times 10^{-19} \text{ C})}{2(1.67 \times 10^{-27} \text{ kg})}\right| \simeq 2.92 \text{ MHz}$$

and this is a radiowave.

7.5 *Theorem:* $\hat{U}_t|\mu = 1/2\rangle = \frac{\hat{U}_t}{\sqrt{2}}(|m = 1/2\rangle + i|m = -1/2\rangle)$ where μ and m are eigenvalues of $\hat{S}_y$ and $\hat{S}_z$ respectively.

Proof. The unit time transform can be removed from both sides, so the proof is just

$$|\mu = 1/2\rangle = \frac{1}{\sqrt{2}}|m = 1/2\rangle + \frac{i}{\sqrt{2}}|m = -1/2\rangle$$

and this can be proven using the change-of-basis matrix from $\text{eig}_\lambda \hat{S}_y$ to $\text{eig}_\lambda \hat{S}_z$. This can be produced using the half-integer spinor rotation operator

$$\hat{R}(\theta, \hat{n}) = \exp\left(-\frac{i\theta}{2}\vec{\sigma} \cdot \hat{n}\right) \quad (4)$$

where θ is the rotation angle, $\hat{n}$ is the axis of rotation (in this case, the x -axis), and $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ is a vector of the Pauli matrices for each axis. For a half-integer spin on the x -axis, this simplifies (via a Taylor expansion not shown) to

$$\hat{R}_x(\theta) = I \cos \frac{\theta}{2} - i\sigma_x \sin \frac{\theta}{2} = \begin{bmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{bmatrix}. \quad (5)$$

Thus, given that we want a counterclockwise rotation from the y -axis to the z -axis, we need to compute $\hat{R}_x(-\pi/2)$, which is

$$\hat{R}_x(-\pi/2) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}$$

and then, taking $|\mu = 1/2\rangle$ to be $(1\ 0)^T$ in the y -basis, we can use $\hat{R}_x(-\pi/2)$ as the change-of-basis matrix from the y -basis to the z -basis as

$$\hat{R}_x(-\pi/2) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix},$$

which, written out, is equivalent to

$$|\mu = 1/2\rangle = \frac{1}{\sqrt{2}} |m = 1/2\rangle + \frac{i}{\sqrt{2}} |m = -1/2\rangle.$$

□

- 7.6 A neutron spinning initially in the $+y$ -direction, ($\mu = 1/2$), spends time t in a z -direction magnetic field of strength B . What is the probability of finding the initial spin ($\mu = 1/2$) at time t ?

This probability boils down to computing $\|\langle \mu = 1/2 | \hat{U}_t | \mu = 1/2 \rangle\|^2$, so all we need to compute is $\langle \mu = 1/2 | \hat{U}_t | \mu = 1/2 \rangle$. First, the time-evolution operator can be expanded to

$$\hat{U}_t = \exp\left(\frac{-it\hat{H}}{\hbar}\right) = \exp\left(\frac{-it\Omega\hat{S}_z}{\hbar}\right)$$

And given that the hamiltonian can only operate on states defined in the z -axis, the time-evolved operator can be expressed using equation 7.13 as

$$U_t \left| \mu = \frac{1}{2} \right\rangle = \frac{1}{\sqrt{2}} \exp\left(\frac{-it\Omega}{2}\right) \left| m = \frac{1}{2} \right\rangle + \frac{i}{\sqrt{2}} \exp\left(\frac{it\Omega}{2}\right) \left| m = -\frac{1}{2} \right\rangle$$

and similarly to the procedure described in the previous problem, we can then calculate $\langle \mu = 1/2 | U_t | \mu = 1/2 \rangle$ by performing a rotation by $\theta = 1/2$ along the x -axis, i.e., we can use $\hat{R}_x(\pi/2)$ as the $z \rightarrow y$ change-of-basis matrix:

$$\hat{R}_x(\pi/2) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix}$$

Thus, expressing $U_t | \mu = 1/2 \rangle$ in the z -basis yields

$$\hat{U}_t | \mu = 1/2 \rangle \equiv \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-it\Omega/2} \\ ie^{it\Omega/2} \end{pmatrix}$$

and if we then perform the rotation into the y -axis using $\hat{R}_x(\theta/2)$

$$\hat{R}_x(\pi/2) \hat{U}_t | \mu = 1/2 \rangle = \frac{1}{2} \begin{pmatrix} \lambda + \lambda^{-1} \\ i(\lambda - \lambda^{-1}) \end{pmatrix}$$

where $\lambda = e^{it\Omega/2}$, or equivalently,

$$\hat{R}_x(\pi/2) \hat{U}_t | \mu = 1/2 \rangle = \begin{pmatrix} \cosh(it\Omega/2) \\ i \sinh(it\Omega/2) \end{pmatrix} = \begin{pmatrix} \cos(t\Omega/2) \\ i \sin(t\Omega/2) \end{pmatrix},$$

and thus, the raw amplitude is

$$\langle \mu = 1/2 | U_t | \mu = 1/2 \rangle = \cos(t\Omega/2)$$

and the probability is

$$|\langle \mu = 1/2 | U_t | \mu = 1/2 \rangle|^2 = \cos^2(t\Omega/2) = \frac{1}{2}(1 + \cos \Omega t)$$

by the double-angle formula.

7.7 *Theorem:* $\exp\left(\frac{-it\hat{H}}{\hbar}\right) = 1 - \frac{it\hat{H}}{\hbar}$ for a stationary state where $E < \hbar/t$.

Proof. This can be proven quickly with a simple Maclaurin expansion. The Maclaurin expansion of e^x is

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \dots,$$

□

so therefore,

$$\exp\left(\frac{-it\hat{H}}{\hbar}\right) = 1 - \frac{it\hat{H}}{\hbar} + \frac{1}{2} \left(\frac{-it\hat{H}}{\hbar}\right)^2 + \dots$$

and as $E < \hbar/t$, then $t\hat{H}/\hbar \ll 1$, so the only two important terms are the first two, meaning that

$$\exp\left(\frac{-it\hat{H}}{\hbar}\right) \simeq 1 - \frac{it\hat{H}}{\hbar}$$

7.8 *Theorem:* $\langle A \rangle = \langle \psi | \hat{A} | \psi \rangle \forall A, \forall \psi$.

Proof.

$$\begin{aligned} \langle \psi | \hat{A} | \psi \rangle &= \sum_A \langle \psi | A \rangle \langle A | \hat{A} | \psi \rangle &&= \sum_A A \langle \psi | A \rangle \langle A | \psi \rangle \\ &= \sum_A A \langle A | \psi \rangle^* \langle A | \psi \rangle &&= \sum_A A |\langle A | \psi \rangle|^2 \\ &= \sum_A A P(A | \psi) &&= \langle A \rangle \end{aligned}$$

□

7.9 Show that

$$\begin{aligned}\hat{U}_t \left| k, \mu = \frac{1}{2} \right\rangle &= \frac{1}{\sqrt{2}} \exp \left(-it \left(\frac{\Omega}{2} + \frac{\hbar k^2}{2m_n} \right) \right) \left| k, m = \frac{1}{2} \right\rangle + \\ &\quad \frac{i}{\sqrt{2}} \exp \left(it \left(\frac{\Omega}{2} - \frac{\hbar k^2}{2m_n} \right) \right) \left| k, m = -\frac{1}{2} \right\rangle\end{aligned}$$

where $\hat{H} = \frac{\hbar^2 \hat{k}^2}{2m_n} + \Omega \hat{S}_z$

Proof.

$$\begin{aligned}\hat{U}_t &= \exp \left(-\frac{it\hat{H}}{\hbar} \right) = \exp \left(-\frac{it}{\hbar} \left[\Omega \hat{S}_z + \frac{\hbar^2 \hat{k}^2}{2m_n} \right] \right) \\ &= \exp \left(-\frac{it}{\hbar} \left[\Omega \hat{m} \hbar + \frac{\hbar^2 \hat{k}^2}{2m_n} \right] \right) = \exp \left(-it \left[\Omega \hat{m} + \frac{\hbar \hat{k}^2}{2m_n} \right] \right),\end{aligned}$$

thus,

$$\begin{aligned}\hat{U}_t \left| k, m = \frac{1}{2} \right\rangle &= \exp \left(-it \left[\frac{\Omega}{2} + \frac{\hbar k^2}{2m_n} \right] \right) \left| k, m = \frac{1}{2} \right\rangle \\ \hat{U}_t \left| k, m = -\frac{1}{2} \right\rangle &= \exp \left(it \left[\frac{\Omega}{2} - \frac{\hbar k^2}{2m_n} \right] \right) \left| k, m = -\frac{1}{2} \right\rangle\end{aligned}$$

and therefore, altogether,

$$\begin{aligned}\hat{U}_t \left| k, \mu = \frac{1}{2} \right\rangle &= \frac{1}{\sqrt{2}} \exp \left(-it \left(\frac{\Omega}{2} + \frac{\hbar k^2}{2m_n} \right) \right) \left| k, m = \frac{1}{2} \right\rangle + \\ &\quad \frac{i}{\sqrt{2}} \exp \left(it \left(\frac{\Omega}{2} - \frac{\hbar k^2}{2m_n} \right) \right) \left| k, m = -\frac{1}{2} \right\rangle\end{aligned}$$

□

7.10 *Theorem:*

$$dk \left| \langle k, \mu = 1/2 | \hat{U}_t | k, \mu = 1/2 \rangle \right|^2 dk = \left| \langle \mu = 1/2 | \hat{U}_t | \mu = 1/2 \rangle \right|^2$$

Proof. Let $\theta = \Omega/2$ and $\phi = \frac{\hbar k^2}{2m_n}$, then

$$\begin{aligned}\hat{U}_t | k, \mu = 1/2 \rangle &= \frac{e^{-it\theta} e^{-it\phi}}{\sqrt{2}} | k, m = 1/2 \rangle + \frac{ie^{it\theta} e^{-it\phi}}{\sqrt{2}} | k, m = -1/2 \rangle \\ &= \frac{e^{-it\phi}}{\sqrt{2}} (e^{-it\theta} | k, m = 1/2 \rangle + ie^{it\theta} | k, m = -1/2 \rangle) \\ &\equiv \frac{e^{-it\phi}}{\sqrt{2}} \begin{pmatrix} e^{-it\theta} \\ ie^{it\theta} \end{pmatrix}\end{aligned}$$

Then, we could compute another $z \rightarrow y$ change-of-basis with $\hat{R}_x(\theta/2)$, but notice how the k -dependent term $e^{-it\phi}$ is a common coefficient to both the $m = 1/2$ and $m = -1/2$ terms. Therefore, it will also be common to the $\mu = 1/2$ and $\mu = -1/2$ terms afterward. Therefore, as $e^{-it\phi}$ is only a phase modifier (as $|e^{i\theta}|^2 = 1$), this term cannot effect the final probability. $\square$

7.12 *Theorem:* $\langle \beta | \hat{A} | \beta \rangle = \langle \alpha | e^{i\hat{\theta}} \hat{A} e^{-i\hat{\theta}} | \alpha \rangle$ where $|\beta\rangle = \hat{U}_t |\alpha\rangle = e^{-i\hat{\theta}} |\alpha\rangle$ and $\hat{\theta} = t\hat{H}/\hbar$.

Proof.

$$\begin{aligned} \langle A \rangle &= \langle \beta | \hat{A} | \beta \rangle &= |\beta\rangle^* \hat{A} |\beta\rangle \\ &= (\hat{U}_t |\alpha\rangle)^* \hat{A} \hat{U}_t |\alpha\rangle &= \langle \alpha | \hat{U}_t^\dagger \hat{A} \hat{U}_t | \alpha \rangle \\ &= \langle \alpha | e^{i\hat{\theta}} \hat{A} e^{-i\hat{\theta}} | \alpha \rangle \end{aligned}$$

$\square$

7.13 *Theorem:*

$$\langle b | \hat{A} | \beta \rangle = \langle a | e^{i\hat{\theta}} \hat{A} e^{-i\hat{\theta}} | \alpha \rangle$$

where $|b\rangle = \hat{U}_t |a\rangle$ and $|\beta\rangle = \hat{U}_t |\alpha\rangle$.

Proof. Follows from the previous problem. Proof is obvious. $\square$